// William Chuang
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Education

University of Arizona M.S. in Mathematics (Expected Spring 2025)

Graduate-level coursework completed under Ph.D. classification.

San Francisco State University M.A. in Mathematics (Spring 2022)

Thesis: The Hausdorff Dimension of Limit Sets of Well-Distributed Schottky Groups.

Link: scholarworks.calstate.edu/downloads/xd07h079g   Advisor: Dr. Chun-Kit Lai

University of San Francisco B.S. in Mathematics; Minor in Computer Science (Fall 2018)

Major GPA: 3.88/4.00. Graduated with Honors.

Research Interests

Interplay between mathematical physics, geometry, and the topology of data. Geometric first-principles frameworks for attention/autoencoders; monodromy/topology of training; automorphic and arithmetic structures in learning; post-quantum cryptography motivated by spectral and symbolic dynamics; curvature-aware optimization and representation.

Documented Historical Firsts (plain-English, to the best of my knowledge)

- First quantum-ready transformer/autoencoder derived from geometric first principles: layers act as Möbius flows with unitary/Hamiltonian evolution.
- Monodromy of neural networks: tracking global model evolution via parameter loops to structure training topology.
- Automorphic attention: attention kernels using Maass–Poincaré and Hecke structures (modular-invariant propagation).
- Langlands-informed training dynamics: arithmetic spectral data guiding update rules at a category level.
- Lorentzian loss-landscape control: using Minkowski isometries to reparameterize non-Euclidean optimization.
- MSIA post-quantum primitive: modular spectral indistinguishability via structured zeta matrices and symbolic dynamics.
- Curved-space neural encryption: bulk–boundary duality (hyperbolic models) for non-invertible encodings at the theory level.

Selected Original Contributions (technical abstracts)

- Quantum-ready transformer/autoencoder architectures from geometric first principles (hyperbolic/PSL(2,R) flows; unitary/Hamiltonian evolution in complex Hilbert spaces).
- Monodromy and loss landscape geometry (topological evolution via parameter loops; Lorentzian reparameterization for non-Euclidean optimization).
- Automorphic kernels in attention (Maass–Poincaré series; Hecke eigenstructures) and Langlands-inspired dualities (arithmetic spectral data informing training dynamics).
- Post-quantum cryptographic constructs grounded in modular/representational spectra with symbolic dynamics and structured zeta assumptions.

Independent Mathematical Theories (concise)

1. **GRAIL**: Geometric Representation Algebra for Intelligent Learning (universal meta-architecture over curved spaces; group-theoretic and categorical operators; supports encryption/symbolic layers).
2. **FPQND**: First-Principle Quantum Neural Dynamics (unitary evolution from Lie/Hamiltonian flows).
3. **LNC**: Langlands–Neural Correspondence (functorial mappings between automorphic and Galois representations to guide training dynamics).
4. **MTD**: Monodromic Training Dynamics (topological descent via parameter loops and covering space dynamics).
5. **CNE**: Curved-Space Neural Encryption (bulk–boundary dualities across admissible GRAIL manifolds).
6. **MSIA**: Modular Spectral Indistinguishability Assumption (post-quantum hardness via symbolic dynamics and zeta matrices).
7. **FCD**: Fractal Cryptographic Dynamics (dimensional/entropic tunability for obfuscation).
8. **ANQFT**: Arithmetic–Neural Quantum Field Theory (mapping loop integrals/periods into neural-algebraic automorphic forms).
9. **CNGR**: Cosmic Neural Galois Representation (motivic/Galois actions and period structures in learnable systems).
10. **VGCF**: Vacuum–Graviton Cascade Formalism (controlled vacuum instabilities with curvature-amplified excitations).
11. **DLYVE**: Dynamic Lee–Yang Vacuum Engineering (criticality in tunable vacuum structures).
12. **QACT**: Quantum Amplification Cascade Theory (operator algebra for amplification cascades with conservation/backreaction controls).

U.S. Law Compliance & Research Context

All research listed is unclassified, produced in academic settings, suitable for public dissemination, and conducted in compliance with U.S. academic norms and export control regimes (e.g., ITAR/EAR public disclosure protocols). No classified work is presented here. Any future collaboration requiring clearance would follow standard eligibility and review procedures.

Other Independent Projects (selected)

University of Arizona: independent study in real/complex analysis and hyperbolic geometry (with Prof. David Glickenstein, Fall 2023); RTG project on scaling factors of self-attention (with Prof. Ning Hao, Fall 2023).

San Francisco State University: computations of Hausdorff dimension of limit sets of Schottky groups (with Dr. Chun-Kit Lai, 2021–2022); projects in topology/combinatorics.

University of San Francisco & Penn State: projects in topology, prime number theorem readings, functional analysis, capstones in graph theory and statistical methods.

Patents & Disclosures (U.S., unclassified)

Unclassified, exploratory academic R&D. Titles shown for identification; no classified work or deployments.

- **Critical Entropy Attention System (CEAS)**
  US Provisional Patent Application No. 63/813,617 (filed May 29, 2025) — AI attention efficiency and stability methods for resource-constrained computation (academic prototype).
- **Analytic Corridor Methods for Critical Synchronization, Memory Integration, and Optimization in Ising-like Neural Systems**
  US Provisional Patent Application No. 63/824,102 (filed Jun 15, 2025) — mathematical frameworks for coordinated memory/optimization in physics-inspired AI (theoretical, unclassified).
- **System and Method for Remote Polygraphy and Internal Voice Capture for Neural Training via Multimodal Electromagnetic Sensing**
  US Provisional Patent Application No. 63/824,053 (filed Jun 15, 2025) — experimental sensing concepts for model training; privacy-preserving by design; unclassified; no operational deployments.
- **Neural Projection Unit with CEAS-Driven Dynamic Beta Scaling and Thermodynamic Synchronization**
  US Patent Application No. 19/232,791 (nonprovisional; filed Jun 9, 2025) — adaptive scaling module to improve convergence/robustness in learning systems (unclassified R&D).
- **Neural Projection Unit with CEAS-Based Dynamic Beta Scaling and Thermodynamic Synchronization**
  US Provisional Patent Application No. 63/820,652 (filed Jun 9, 2025) — related provisional design supporting dynamic parameter scaling (academic research only).
- **The Modular Symbolic Intelligence Architecture (MSIA)**
  US Provisional Patent Application No. 63/809,257 (filed May 20, 2025) — hybrid symbolic–neural

architecture for interpretable, mission-relevant AI (unclassified, exploratory).

- **Hyperbolic Autoencoder and Transformer Framework for Real-Time Landscape Modification and Spacetime Adaptation**
  US Provisional Patent Application No. 63/773,441 (filed Mar 18, 2025) — geometric deep learning for real-time adaptive modeling; working prototype; unclassified academic research.

Pre-Baccalaureate Independent Projects (historical, unclassified)

Research student at NTU LeCosPA (2011–2013) with seminar topics in QFT, gravity, holography, critical phenomena, and analog models; independent readings in GR and quantum mechanics at NTU/NDHU under faculty supervision. Academic in nature; no foreign government employment; no classified or restricted work.
Old Profile: Tao-Mao Chuang
LeCosPA People (Archived)

Work Experience (Teaching & Research)

University of Arizona (2022–2025): Graduate Teaching Assistant (College Algebra; Calculus I), Grader/Tutor (Calculus I/II), common final grading; teaching mentors/advisors include Mitchell Wilson, Tina Deemer, Catherine Yslas, Oussama Ben Said, Tynan Lazarus, and Prof. David Glickenstein.

San Francisco State University (Spring 2022): Graduate Teaching Assistant; Instructor (fourth-hour components for Calculus I/II); Grader (Calculus II).

Other Academic Experience

San Francisco State University: Graduate TA (Pre-Calculus, Fall 2019).

University of San Francisco: San Francisco Math Circle (Fall 2016).

National Dong Hwa University: Undergraduate research assistant and tutoring in Calculus/General Physics (2008–2010).

Awards and Honors

MSRI Summer Graduate School on Metric Geometry and Geometric Analysis (Oxford nomination, 2021).

Dean's Honor Roll (USF: multiple terms).

MASS Program Scholarship (Penn State, Fall 2017; covered tuition/fees).

Pi Mu Epsilon Honor Society.

Admitted to select theoretical physics schools/workshops (NTU, NTHU, Academia Sinica, 2009–2012).

Certificates

Protecting God's Children Online Awareness 4.0 (Order of Malta, Western Association, Mar 7, 2025).

Information Security Awareness (UA, Aug 27, 2023). Safety Preparedness (UA, Dec 8, 2023).

MASS Program Completion (Penn State, 2017). Recognition of Service Award (ACM SIGMOD, 2016).

Tackling the Challenges of Big Data (MIT CSAIL online, Feb–Mar 2015).

Service to the Community

Long-term service beginning in 1998 (altar server at Fu Jen Catholic University). High school hospital service in Taipei (elderly and children with disabilities). University and adult service includes the Tucson Math Circle (site support), Poverello House shifts, and parish blood drives. Future plans include expanded service with the Order of Malta, Knights of Columbus, Catholic Action in San Diego, and related initiatives.

Skills

Problem solving; Python, C/C++, Java, R, Lisp, Shell (sed/awk); LaTeX, Mathematica; PyTorch, Lightning, NumPy, Pandas, scikit-learn, Matplotlib; algorithm design for theoretical prototyping.

Memberships & Affiliations

Order of Malta — Auxiliary Member (Arizona Area), 2025–present.

Knights of Columbus — Fourth Degree (Sir Knight), 2023–present.

Space Force Association — Member, 2025–present.